

Phase 6 - Visualization and Viewable Ready Output

Phase 6 turns the numerical results from Phases 1 through 5 into the figures, maps, and LaTeX table fragments that appear in the thesis. The phase produces approximately 30 publication-ready outputs across two computational stacks: **R** handles all spatial and cartographic output (national choropleths, county-level opportunity cost maps, Oahu micro-maps, bivariate cost-versus-density maps, and the zoning intersection map), and **Julia** handles all non-spatial output (forest plots, MICE density convergence diagnostics, marginal effects charts, raincloud diagnostics, the Hawaii Gap dumbbell, the Lorenz curve, the Preservation Paradox waffle chart, the counterfactual area comparison, and the LaTeX table fragments for direct `\input{}` into the thesis source).

I deliberately do not use Python in Phase 6. After comparison runs across all three candidate languages, R's `sf` + `ggplot2` + `tigris` stack produced consistently better cartographic output than `geopandas` + `matplotlib`, and Julia's `CairoMakie` produced cleaner statistical plots than Python's `matplotlib` or `seaborn` for the chart types this thesis requires. I therefore break the strict tri-language parity rule that governs Phases 1 through 5, in favor of producing better-looking output. The numerical inputs to Phase 6 are still tri-language: each chart that displays per-language results does so by reading the Python, R, and Julia outputs from earlier phases independently, with the Grand Mean computed as the arithmetic mean across the three Rubin-pooled estimates.

The phase is organized as two master scripts (`Phase_6.R` and `Phase_6.jl`) that orchestrate seven R modules and seven Julia modules respectively. Each module is self-contained and produces its own labeled output set under `Final_Thesis_Figures/`. The figure naming convention encodes both the chart number and the version (`5_141_Forest_Plot_Combined.png` for Forest Plot version 1.41, `12_141_Zoning_Waffle_Chart_TriLanguage.png` for Zoning Waffle Chart version 1.41), which lets the thesis prose reference figures stably even as visualization choices iterate.

Key result

The headline figures generated by Phase 6 are the load-bearing visual evidence of the thesis:

- **National opportunity cost maps** (county-level, both Grand Mean and observed-only, plus state-level versions of each) make the spatial concentration in California, Florida, New York, Texas, and Hawaii immediately legible.
- **Forest plot of regression coefficients** displays Python, R, and Julia coefficients side by side with 95% confidence intervals, demonstrating tri-language convergence to within visual indistinguishability for the Holes coefficient and the urban indicator.
- **MICE density convergence sequence** ($n=20 \rightarrow 40 \rightarrow 60 \rightarrow 80 \rightarrow 100$) shows the imputed acreage distribution converging on the observed parcel distribution as $\$M$ increases, justifying the choice of $\$M = 100\$$ visually rather than just computationally.
- **Hawaii Gap dumbbell chart** displays each Oahu course as a horizontal segment from its agricultural-floor value ($\text{USDA} \times \text{acreage}$) to its HBU model value ($\text{FHFA} \times \text{acreage}$, Rubin-pooled), making the per-course zoning tax visible as a length on the chart.
- **Preservation Paradox waffle chart** decomposes the $\$28.6\text{B}$ Oahu opportunity cost by zoning class, showing visually that the bulk of the modeled value sits in zones where redevelopment is statutorily prohibited.

- **Lorenz curve** of per-course opportunity cost shows the high inequality of the distribution: a small share of urban courses dominates the national aggregate.

Three LaTeX table fragments ([Table1_Acreage](#), [Table2_Regression](#), [Table3_Hawaii_Geo](#)) are also generated by Julia in this phase and are dropped directly into the thesis source via `\input{}`, eliminating any manual transcription step between the analytical outputs and the published document.

What was solved

The most consequential standardization decision I made in Phase 6 was that the Grand Mean across the three languages is computed as the arithmetic mean of three independently Rubin-pooled estimates, not as a Rubin pool over the combined 300 imputed datasets. The two procedures are not equivalent: pooling Rubin's Rules across all 300 imputations would treat the three languages as drawing from the same underlying imputation distribution, which they are not. Each language uses a different MICE backend (LightGBM in Python, Random Forest in R, [Mice.jl](#) in Julia), and the Grand Mean is intentionally an *across-implementation* average that demonstrates robustness to backend choice. I propagate this methodological clarification through all charts and tables that display the Grand Mean alongside per-language Rubin pools.

A persistent Map 15.1 issue (log-residual choropleth rendering all-gray due to NaN log-residual range) is documented but not blocking - the dollar-residual map is unaffected and is the version used in the thesis. The bug is a pre-existing computation issue rather than a Phase 6 regression.

Outputs from Phase 6 feed directly into the thesis manuscript, bypassing Phase 7's traditional role as a separate "documentation" stage. The thesis figures and tables in Section 5 are not reproductions of Phase 6 output; they are exactly the files Phase 6 generates.